Zedong Peng

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Education

M.Eng. in Electronic and Information Engineering, School of Computer Science, *Shanghai Jiao Tong University* — Shanghai, China, 2024 – 2027 (expected)

Advised by Prof. Jieru Zhao and Prof. Qiang Xu. Transcript

B.Eng. in Information Engineering, School of Electronic Information and Electrical Engineering, *Shanghai Jiao Tong University* — Shanghai, China, 2020 – 2024

Outstanding Graduate. Transcript

Experience

Research Intern, *National Center of Technology Innovation for EDA* — Nanjing, China, July 2024 – 2025

Advised by Prof. Qiang Xu and Prof. Jieru Zhao.

- The HLS part of LLM4EDA: LLM post-training and data construction for a domain-specific language (blog post).

Machine Learning Intern, *Intel* — Shanghai, China, Nov 2023 – July 2024

Advised by Yuan Zhou.

- Spark SQL native engine Gluten: optimizing the CI pipeline.
- LLM framework optimization: Intel device support for vLLM.
- LLM applications: text to SQL.

Winter School, *Data Science Institute, Imperial College London* — London, UK, Jan 2024 – Feb 2024

- Applied vector representations to analyze language features in COVID-19 text.

Research Intern, *Shanghai Jiao Tong University* — Shanghai, China, July 2022 – July 2024

Advised by Prof. Liyao Xiang.

- Security-focused machine learning research.

Recent Work

MemoryTax: The Bitter Lesson of Agent Memory — Oct 2026

Does agent memory still help once the model can search the raw history itself? Memory systems and coding agents on LoCoMo, with three models.

llama.cpp on an FPGA — In progress

A llama.cpp backend that runs LLM inference on an AMD Alveo U280 FPGA card; twelve models from four families run on it. The code is private for now.

Publications

DiffHLS: Differential Learning for High-Level Synthesis QoR Prediction with GNNs and LLM Code Embeddings — 2026

Zedong Peng, Zeju Li, Qiang Xu, Jieru Zhao

IEEE/ACM International Symposium of EDA (ISEDA)

HIPPO: A Hierarchy-Preserving and Noise-Tolerant Pre-HLS Power Modeling Framework for FPGA 2025
Zefan Lin, **Zedong Peng**, Mingzhe Gao, Jieru Zhao, Zhe Lin
Proceedings of the 44th IEEE/ACM International Conference on Computer-Aided Design (ICCAD)

ForgeHLS: A Large-Scale, Open-Source Dataset for High-Level Synthesis 2025
Zedong Peng, Zeju Li, Mingzhe Gao, Qiang Xu, Chen Zhang, Jieru Zhao
arXiv preprint arXiv:2507.03255

DeepCircuitX: A Comprehensive Repository-Level Dataset for RTL Code Understanding, Generation, and PPA Analysis 2025
Zeju Li, Changran Xu, Zhengyuan Shi, **Zedong Peng**, Yi Liu, Yunhao Zhou, Lingfeng Zhou, Chengyu Ma, Jianyuan Zhong, Xi Wang, Jieru Zhao, Zhufei Chu, Xiaoyan Yang, Qiang Xu
IEEE International Conference on LLM-Aided Design (ICLAD)

A verifiable and privacy-preserving federated learning training framework 2024
Haohua Duan, **Zedong Peng**, Liyao Xiang, Yuncong Hu, Bo Li
IEEE Transactions on Dependable and Secure Computing

Honors and Awards

First-class Postgraduate Academic Scholarship, Shanghai Jiao Tong University 2025
Outstanding Graduate, Shanghai Jiao Tong University 2024
2nd Prize, 13th National University Mathematics Competition 2021
1st Prize, SJTU “Yunhan Cup” RobotMaster Intramural Competition 2021
Liu Yongling Foundation Scholarship 2021
Advanced Individual in Social Practice, Shanghai Jiao Tong University 2021

Projects

Robotic Dog Development Shanghai, China
Advised by Prof. Weiming Zhao and Xiaomi Technology. Nov 2023 – Nov 2024
- Implemented algorithms that improved locomotion control and user interaction.

VR Game “Strange Zoo” Shanghai, China
Original VR adventure puzzle game for the course CS085, VR & AR Design Practice. Sept 2021 – Jan 2022
- Built interactive gameplay mechanics in Unity3D with C# and Microsoft VR tooling.

Robot Cart Design and Control Shanghai, China
Champion project of the SJTU “Yunhan Cup” RobotMaster Intramural Competition. Sept 2021 – Jan 2022
- Designed and built a competition robot with Arduino-based control; won 1st prize.

Skills

Programming: C++, Python, SQL, Verilog, Shell, Docker, MATLAB
Languages: Chinese (native), English (CET-6, 481), German (A1)

Updated October 2026